

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I P. HEMANTH REDDY (192312173) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)

Role of the OSI Model in Software-Defined Networking (SDN)

CSA0716 computer networks

Presented by: M.M.ANIL
P.HEMANTH REDDY
U.PAVANESH REDDY
RAHUL REDDY

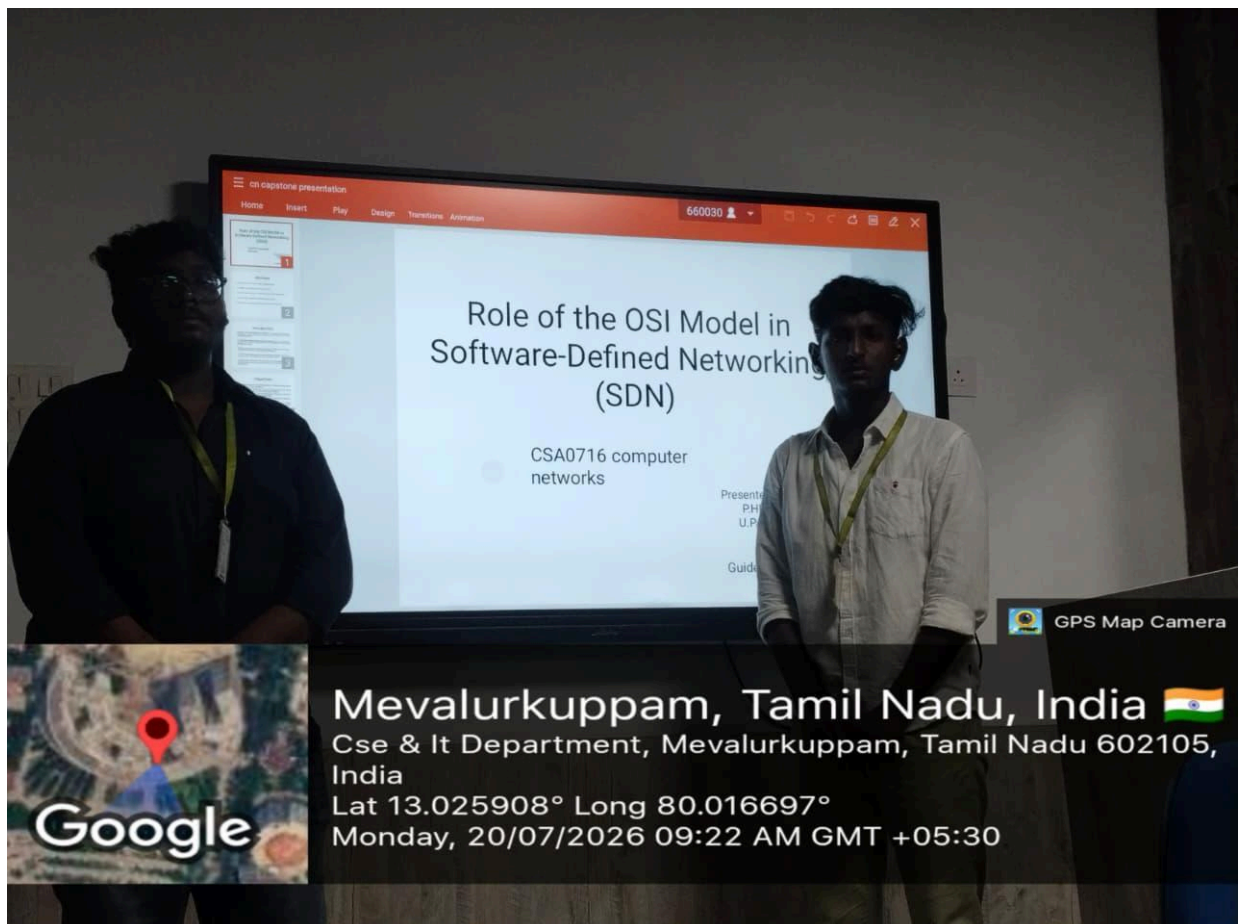
Guided by: P.ABERNA

Abstract

- SDN separates the control plane and data plane.
- It enables centralized network management.
- The OSI model provides a layered communication framework.
- Each OSI layer supports SDN network operations.
- SDN improves flexibility, scalability, and network efficiency.

Introduction

- **Software-Defined Networking (SDN)** is an innovative networking architecture that separates the network control functions from the forwarding functions.
- The **OSI (Open Systems Interconnection) Model** provides a standardized framework for understanding how data is transmitted across a network through seven layers.
- SDN uses a centralized controller to manage and configure network devices, making network operations more flexible and efficient.
- The OSI model helps explain the interaction between SDN components, protocols, and communication processes at different layers.
- By combining the principles of the OSI model with SDN technology, modern networks achieve improved scalability, automation, performance, and easier management.



RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginning
Understandi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technolo gies	Good understandi ng with proper integration of most concepts	Basic understandi ng; limited or partial integration	Poor understandi ng; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementat ion with minor issues	Basic implementat ion with errors or inefficiencie s	Incorrect or incomplete implementat ion
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate	Good analysis	Basic analysis	Poor or incorrect

		interpretation and validation of results	with reasonable interpretation	with limited insights	analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation